

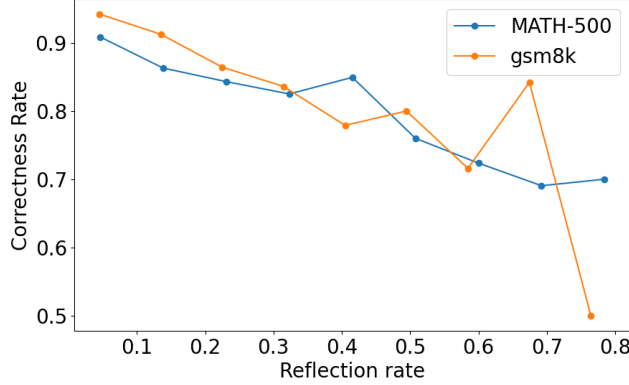


Figure 5: Relationship between correctness rate and reflection rate on MATH-500 and GSM8k datasets. Higher reflection frequency correlates with lower accuracy, partly due to more reflections are generated on difficult questions.

Category	Model	Metric	Reflection Strength			
			-0.96	-0.48	0	0.48
GSM-8k	DS-Llama-8b	Accuracy	88.34%	89.46%	90.09%	90.25%
		Tokens	821.0	1032.6	1595.7	3577.1
	QwQ-32b	Accuracy	96.36%	96.50%	96.50%	96.44%
		Tokens	1006.7	1162.5	1488.6	2256.9
	DS-qwen-14b	Accuracy	95.07%	95.15%	95.15%	94.84%
		Tokens	747.8	880.2	1315.9	3746.4
MATH-500	DS-Llama-8b	Accuracy	82.14%	84.46%	85.98%	86.90%
		Tokens	2738.1	3123.8	4000.7	6017.8
	QwQ-32b	Accuracy	92.72%	92.58%	93.06%	93.08%
		Tokens	2992.9	3253.4	3786.0	5028.9
	DS-qwen-14b	Accuracy	89.22%	90.18%	91.44%	91.86%
		Tokens	2247.1	2534.7	3315.3	5789.0

Table 2: **Accuracy and average reasoning token usage under different intervention strengths.** Results are reported on GSM8k and MATH-500 datasets for DeepSeek-R1 Llama 8B, and DeepSeek-R1 Qwen 14B, and QwQ-32B. The numbers reported are averaged over 10 runs. Negative intervention strengths reduce reflection frequency and reasoning token usage with minimal accuracy loss, suggesting potential reflection redundancy.

non-distilled model. The results show that in both cases, interventions effectively control the number of reflection steps. In terms of accuracy, DeepSeek-R1 Llama 8b gains marginal improvement with more reflections, while QwQ-32b is largely insensitive to the number of reflections. We observe:

1. Most models are less sensitive to additional reflections. From the table, the only model that benefits from positive intervention (more reflections) is the DeepSeek-Llama-8b distilled model, which receives 0.16% and 0.92% accuracy gain on GSM8k and MATH-500, respectively, at the cost of around 2000 additional reasoning tokens for each question.
2. Reflection redundancy exists in many cases, especially for stronger models. For example, in QwQ-32b model, the largest model in our test, the performance loss is only 0.14% and 0.34% on two datasets at intervention strength -0.96 , while the reasoning token budget is reduced by 32.4% and 21.0%, respectively. This demonstrates that reasoning cost, which is substantially higher than non-reasoning models, can be reduced with minimal accuracy loss.

To further understand how reflection affects the trade-off between thinking cost and performance, we calculate the reasoning token usage and accuracy under different intervention strengths and report the results in table 2. For each question, we sample 10 responses and report the mean result. The results confirm our findings, showing that in many cases, reflections can be reduced without sacrificing performance.

Category	Model	Metric	Reflection Strength			
			-0.96	-0.48	0	0.48
Professional accounting	DS-Llama-8b	Accuracy	50.1%	53.4%	56.5%	57.3%
		Tokens	1453.6	1668.8	2097.5	2807.7
	DS-qwen-14b	Accuracy	78.5%	76.8%	77.8%	77.6%
		Tokens	983.9	1103.1	1482.1	2470.1
	QwQ-32b	Accuracy	89.3%	89.5%	88.5%	89.2%
		Tokens	1231.2	1313.7	1648.0	2234.3
Highschool computer science	DS-Llama-8b	Accuracy	79.6%	82.7%	87.3%	88.0%
		Tokens	1016.1	1157.9	1365.4	1970.4
	DS-qwen-14b	Accuracy	95.2%	95.4%	95.0%	94.8%
		Tokens	711.9	787.9	933.5	1498.7
	QwQ-32b	Accuracy	96.6%	96.2%	96.7%	97.0%
		Tokens	771.6	741.9	871.0	1004.7
Formal logic	DS-Llama-8b	Accuracy	60.5%	61.0%	62.1%	62.7%
		Tokens	2266.5	2586.9	3378.3	4553.5
	DS-qwen-14b	Accuracy	91.8%	92.2%	92.6%	92.8%
		Tokens	1287.2	1440.1	1891.4	3196.5
	QwQ-32b	Accuracy	96.3%	95.5%	95.7%	96.0%
		Tokens	1481.4	1447.8	1716.6	2175.6

Table 3: **Accuracy and reasoning token usage under different intervention strengths(λ) on MMLU subsets.** The numbers reported are averaged over 10 runs. Smaller models (e.g., DS-Llama-8B) are more sensitive to reflection reduction, while larger models (DS-Qwen-14B and QwQ-32B) maintain accuracy with fewer reflections, saving up to 33.6% of reasoning tokens.

To evaluate the effectiveness of ReflCtrl, we compare it with the baseline, NoWait [Wang et al., 2025a]. NoWait is a recent work that reduces redundant reflection by directly suppressing corresponding reflection tokens. We plot the accuracy versus number of thinking tokens in Fig. 3. From the results, we can see that ReflCtrl is more flexible: the intervention strength can control the trade-off between performance and cost, while NoWait can only completely disable reflection. Additionally, ReflCtrl generally incurs smaller performance loss under similar token budget.

4.3 Stepwise steering

In this section, we study the stepwise steering strategy for controlling model reflection. We compare it with a baseline method where the intervention is applied to all generation tokens. In this experiment, we use DeepSeek-R1 Llama 8b as the base model and evaluate on GSM8k dataset. As shown in figs. 4a and 4b, we observe that:

1. Under the same intervention strength, stepwise intervention achieves performance similar to intervention at all tokens. The baseline method produces stronger effects when applying positive intervention, i.e. increasing model’s reflection.
2. In terms of accuracy, stepwise intervention maintains accuracy close to the original model, whereas the baseline method degrades performance significantly at larger intervention strengths (< -0.2 or > 0.3). Fig. 4b further shows that, under the same thinking token usage, stepwise intervention generally achieves higher accuracy.

4.4 Ablation study: impact of layers

In this section, we study the effect of applying intervention at different layers of the LLM. We experiment with two strategies: Skipping the first k layers and skipping the last k layers. The results in Fig. 6 indicate that the performance is best when skipping both the first and last six layers. We adopt this configuration accordingly in our main experiments.

4.5 Non-mathematical benchmarks

To evaluate the proposed ReflCtrl on non-mathematical reasoning tasks, we conduct extensive experiments on MMLU subsets. As shown in table 3, we observe a similar phenomenon as in math tasks. In general, the smallest model, DeepSeek Llama 8b, is most sensitive to reflection. In contrast, the larger DeepSeek Qwen 14b and QwQ-32b are hardly affected by the reduction of reflections. Using the proposed stepwise steering, up to 33.6% of reasoning tokens can be saved.

5 Conclusion and limitation

In this work, we propose ReflCtrl, a representation engineering framework for understanding and steering self-reflection behavior in reasoning LLMs. By segmenting the model’s reasoning into thinking steps and identifying reflection-related steps, we extract a reflection direction in the latent space, enabling direct control over self-reflection frequency during inference. We further introduce a **stepwise steering** strategy that only applies interventions at the start of new thinking steps, substantially reducing reasoning token usage while preserving performance. Across multiple math and general-domain reasoning benchmarks, we find that:

1. Reflection redundancy is common, particularly in stronger models where minimal accuracy loss is observed when reflections are reduced.
2. Reflection direction is correlated with internal uncertainty signals, implying that the reflection behavior may be controlled by model’s internal uncertainty perception.
3. Stepwise steering can largely mitigate performance loss. Compared with token-level intervention, stepwise steering can achieve similar intervention performance with over 5% accuracy improvement.

6 Limitations

Despite these promising results, our work also has some limitations. First, the identification of reasoning steps relies on keyword search, which may be model specific since different models could prefer different reflection cues. Second, our ReflCtrl only works for open-source models and it remains unclear whether it generalizes to SOTA closed-source models such as GPT-4 or Claude, which is a shared limitation of representation engineering methods.

For future work, we believe that developing uncertainty-aware dynamic steering is a promising direction: our results preliminarily show a connection between uncertainty and self-reflection. The current steering method applies a fixed strength across all questions and throughout the generation process. Enabling the model to dynamically adjust steering strength during inference could substantially improve reflection efficiency and further reduce inference cost in reasoning LLMs.

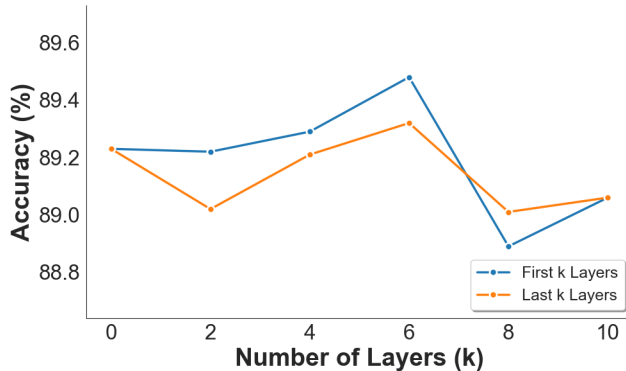


Figure 6: **Effect of applying interventions to different layers of the LLM.** We vary the number of skipped layers at the bottom and top of the network, with intervention strength fixed at $\lambda = -0.48$. Accuracy is highest when skipping the first and last six layers, which we adopt as the default configuration in our main experiments.